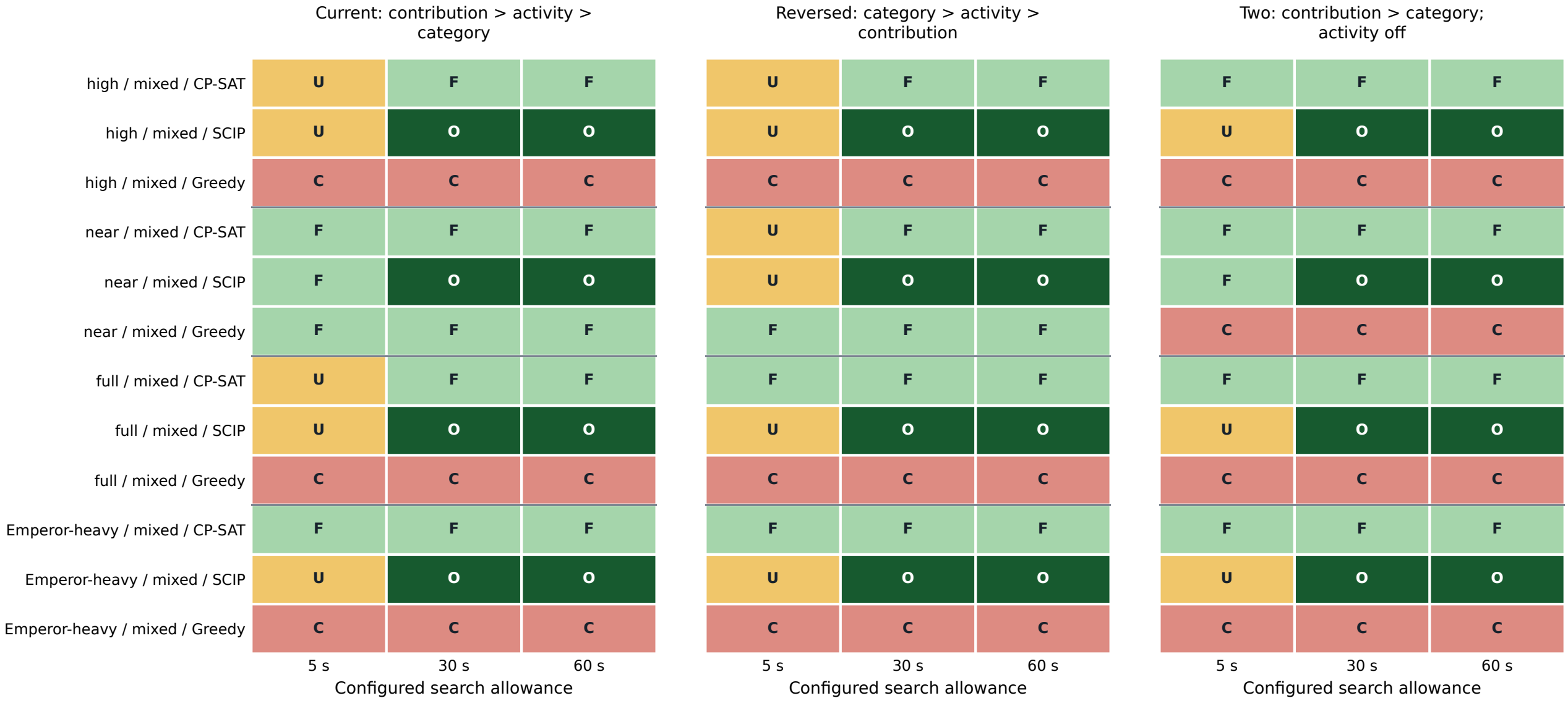
Study A | Complete primary cohort: five-second search allowance

	Current: contribution > activity > category			Reversed: category > activity > contribution			Two: contribution > category; activity off			All ordinary preferences disabled		
	CP-SAT	SCIP	Greedy	CP-SAT	SCIP	Greedy	CP-SAT	SCIP	Greedy	CP-SAT	SCIP	Greedy
small / mixed	O	O	F	O	O	C	O	O	F	O	O	C
small / conflict	O	O	F	O	O	C	O	O	F	O	O	C
small / tied	O	O	C	O	O	C	O	O	C	O	O	C
small / accessible	O	O	F	O	O	C	O	O	F	O	O	C
typical / mixed	F	O	C	F	O	C	F	O	C	O	O	C
typical / conflict	F	O	C	F	O	C	F	O	C	O	O	C
typical / tied	O	U	C	O	U	C	F	U	C	O	U	C
typical / accessible	F	O	C	F	O	C	F	O	C	O	O	C
high / mixed	U	U	C	U	U	C	F	U	C	O	O	C
high / conflict	F	U	C	F	U	C	F	U	C	O	O	C
high / tied	U	U	C	U	U	C	F	U	C	O	U	C
high / accessible	F	U	C	F	U	C	F	U	C	U	O	C
near / mixed	F	F	F	U	U	F	F	F	C	O	O	C
near / conflict	U	U	F	F	U	F	F	F	C	O	O	C
near / tied	U	U	C	U	U	C	U	U	C	O	U	C
near / accessible	F	F	C	F	U	C	F	F	C	O	O	C
Emperor-heavy / mixed	F	U	C	F	U	C	F	U	C	O	O	C
full / mixed	U	U	C	F	U	C	F	U	C	O	O	C
full / conflict	F	U	C	F	U	C	F	U	C	O	O	C
full / tied	U	U	C	U	U	C	U	U	C	O	U	C
full / accessible	F	U	F	F	U	F	F	U	F	O	O	C

O: validated optimal
F: validated feasible
U: no validated map
C: construction failure
I: invalid output
E: incomplete execution

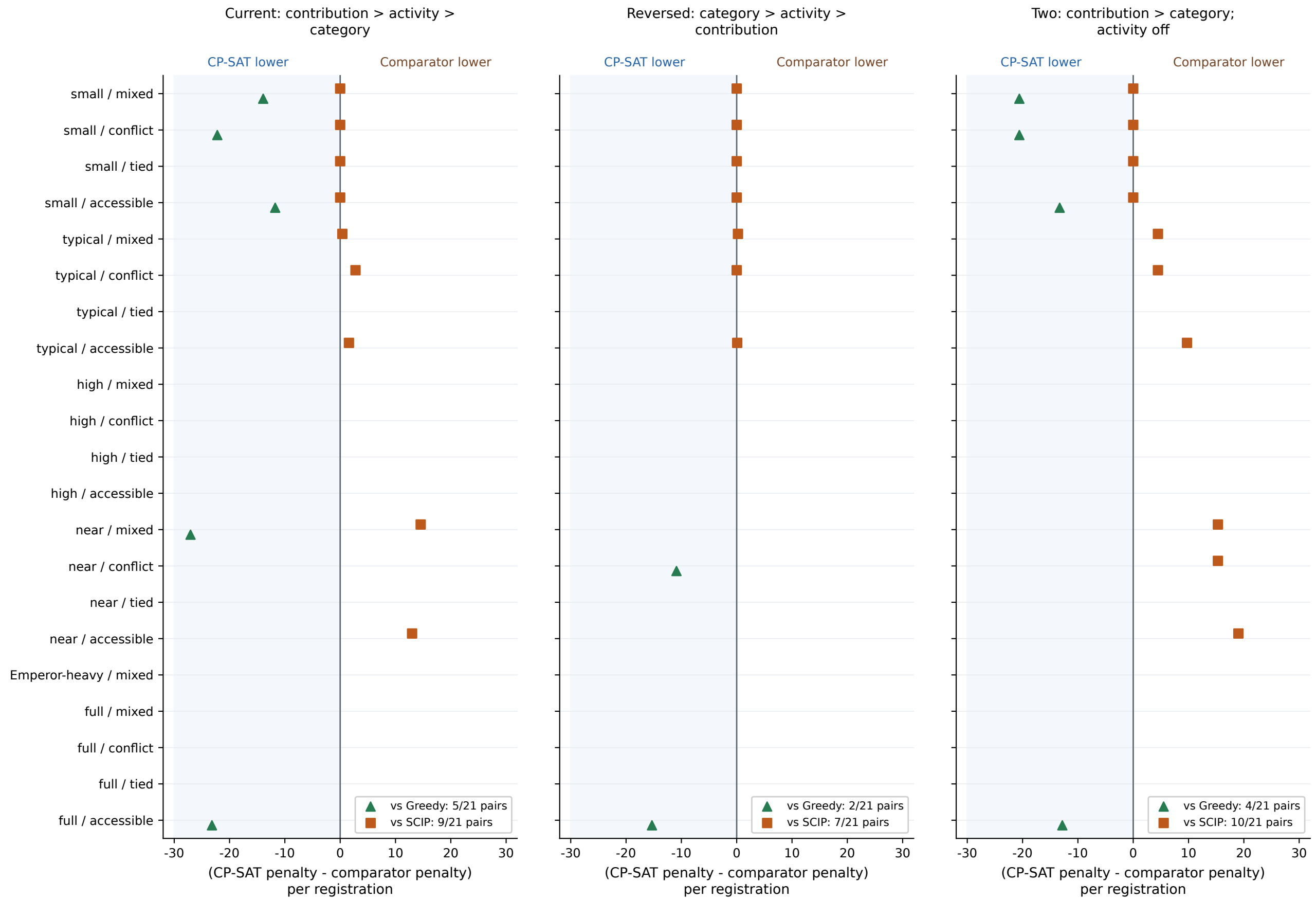
21 events per setting; one worker; no canonicalization. Letters preserve outcome distinctions in grayscale. No invalid outputs or incomplete scheduled executions occurred. Disabled preferences are a feasibility control.

Study A | Matched outcomes as search allowance increases



Only four predefined large mixed-profile events and three active preference settings were tested at every allowance. Each column is a fresh run.
Other events and the disabled setting were not tested at 30/60 seconds.

Study A | Paired preference penalties at five-second search



Zero means equal modeled penalties, not zero-cost seating or a presumed optimum. Blank rows have no both-valid pair. CP-SAT is the subtraction reference only. Settings use different weights and must not be pooled into one quality score.

Study A | Fresh-run budget endpoints: availability, quality and elapsed time

Current: contribution > activity > category

Reversed: category > activity > contribution

Two: contribution > category; activity off

Validated maps / 4 events

Paired penalty change per registration

Observed allocation total (s)

Configured search allowance (s)

5 to 30 s

30 to 60 s

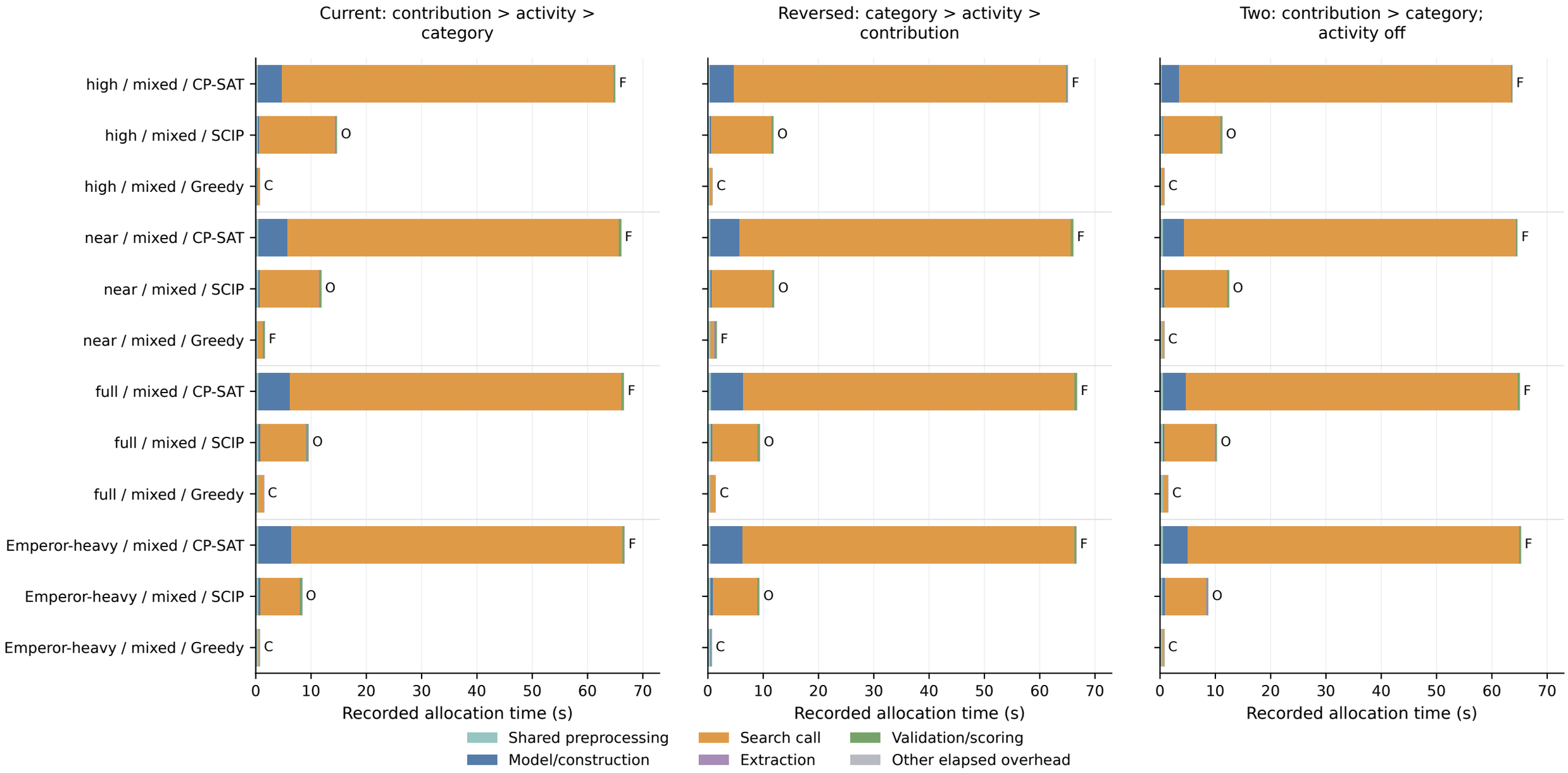
CP-SAT

SCIP

Greedy

Same four large mixed events; one worker; no canonicalization. Middle = (later - earlier) penalty / registrations, both-valid only; negative means lower penalty. Bottom row includes all executions; x marks no validated map. Endpoints are not a continuous solve trajectory or a time-to-first-solution trace.

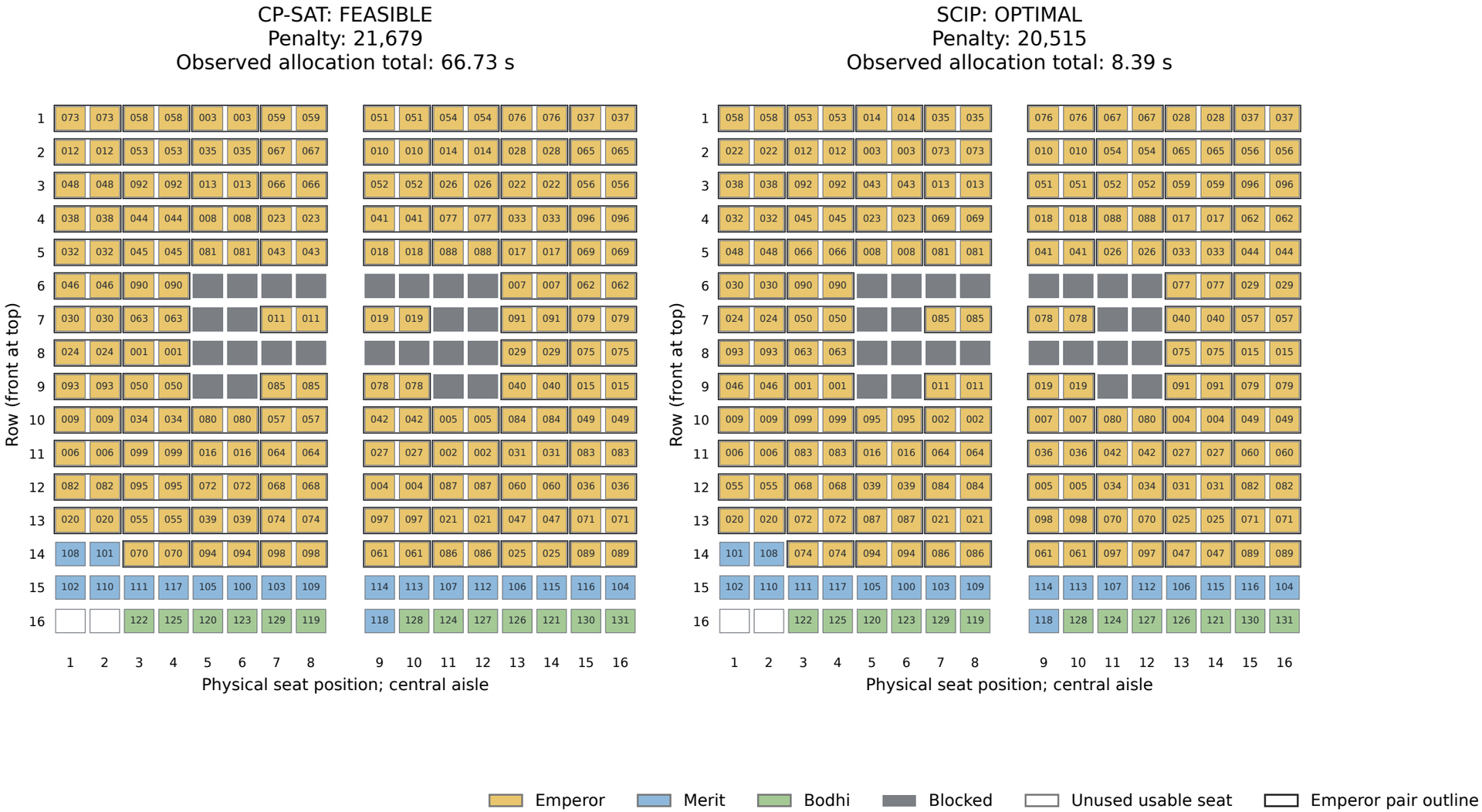
Study A | Where allocation time was spent at the 60-second search allowance



Each bar is one saved execution, not an average. O = validated optimal; F = validated feasible; C = construction failure. Native presolve is inside the search call; these measurements do not isolate a SCIP startup phase. No service/network timing is included.

Study A | The 230-seat event at a common 60-second search allowance

99 Emperor registrations (198 seats), 19 Merit and 13 Bodhi; 131 registration units; current preference order



Identical controlled settings: one worker; no canonicalization; fresh 60-second searches. CP-SAT remained feasible without proof; SCIP proved the minimum. Greedy returned no map. Registration numbers identify placements, not preference quality.